

USA Math Olympiad 1976 — P5/5

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The polynomials $A(x)$, $B(x)$, $C(x)$ and $D(x)$ satisfy the equation

$$A(x^5) + xB(x^5) + x^2C(x^5) = (1 + x + x^2 + x^3 + x^4)D(x)$$

Show that $A(1) = 0$.

Solution

Observe the following

$$\begin{aligned} A(x^5) + xB(x^5) + x^2C(x^5) &= \frac{x^5 - 1}{x - 1} D(x) \\ (x - 1)[A(x^5) + xB(x^5) + x^2C(x^5)] &= (x^5 - 1)D(x) \end{aligned}$$

Take a 5th root of unity z that isn't 1, so $z = e^{2\pi i k/5}$ for $k = 1, 2, 3, 4$. Then

$$p(z) := A(1) + zB(1) + z^2C(1) = 0$$

has $4 > \deg p = 2$ roots which implies that $p(z) = 0$ for all $z \in \mathbb{C}$. Then since $p(0) = 0$ we have $A(1) = 0$. ■